TSET Health Promotion Research Center

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Do smokers initially unmotivated to quit open smartphone messages that promote and support smoking cessation attempts?

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Background

- Intervention messages that focus on reducing cancer risk behaviors (e.g., smoking) may be prompted and delivered via smartphones.[1,2]
- Compliance with viewing such messages may directly impact the efficacy of smartphone-based interventions.
- Prior research suggests compliance varies across demographics and declines over time.[3,4]
- Yet, no research has investigated predictors of varying rates of decline in message compliance across time.

Objectives

- Examine demographic, mental health, and message type predictors of actually viewing prompted messages.
- Identify predictors of declining message compliance over time.

Methods

- A three-group randomized controlled trial enrolled smokers unmotivated to quit within next 30 days (***N*=152**).
- For 182 days, 2 messages delivered/day that were either:
  1. Tailored to cessation stage.
  2. Fact/trivia unrelated to smoking.
- Compliance: percentage of messages opened/viewed.
- Predictors: message type, age, sex, race, geographic region, employment, education, any mental health diagnosis.

Analyses

- Linear regression models correlated predictors to aggregated message compliance.
- Linear mixed effects model plotted compliance across weeks in the study with predictors as moderators.

Table 1: Participant Demographics and Message Compliance

Variable	*n* (%)	Variable	M±SD or *n* (%)
Female	103 (67.8%)	**Age**	50.0±12.5
Race		**Compliance**	
White	113 (74.3%)	Morning Messages	103.0 (60.4%)
Black	13 (8.6%)	Evening Messages	104.3 (60.9%)
American Indian	20 (13.1%)	Total Messages	207.3 (60.6%)
Other	6 (4.0%)		
Employed	72 (47.4%)	**Urban**	117 (77.0%)
Any Mental Health Diagnosis	93 (61.2%)	**Education** (years)	13.3±1.6

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Results

- 31,503 messages were viewed across a total of 17,580 days.
- Participants viewed 60.6% of all prompted messages.

Linear Regression Modeling

- Age was the only significant predictor of compliance.
- Age was positively associated with total and morning message compliance (p=0.018 and 0.007) and marginally related to evening message compliance (p=0.053).

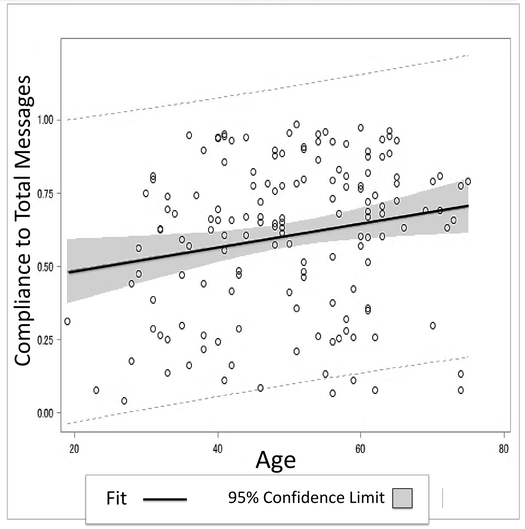


Figure 1. Linear Regression Model: Age and Aggregated Message Compliance.

Table 2: Linear Mixed Effects Model: Interactions Between Predictors and Weeks in the Study on Message Compliance

Variable	Modalities	Morning		Evening		Total	
		β	Sig.	β	Sig.	β	Sig.
Message Type	Tailored Messages Fact/Trivia Messages						
Gender	Female Male	-0.0079 -0.0183	****	-0.0083 -0.0171	****	-0.0081 -0.0177	****
Race/Ethnicity	White Other						
Geographic Region	Urban Rural	-0.0124 -0.0078	***			-0.012 -0.0086	**
Employment	Employed Unemployed	-0.0128 -0.0099	**			-0.0123 -0.0103	*
Education	Years	0.0009	**	0.0017	****	0.0013	****
Any Mental Health Diagnosis	Yes No						

Blank $p > 0.05$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

Linear Mixed Effects Modeling

- Number of weeks in the study was associated with a decline in morning (β=-0.011, p<0.001), evening (β=-0.011, p<0.001) and total (β=-0.011, p<0.001) messages viewed.
- For each modality, the β indicates the rate at which message compliance decreased from one week to the next (see Table 2).
- Linear mixed effects modeling excluded age due to strong collinearity (r=0.89) between weeks in study and the interaction term of weeks in study and age.

Results (cont.)

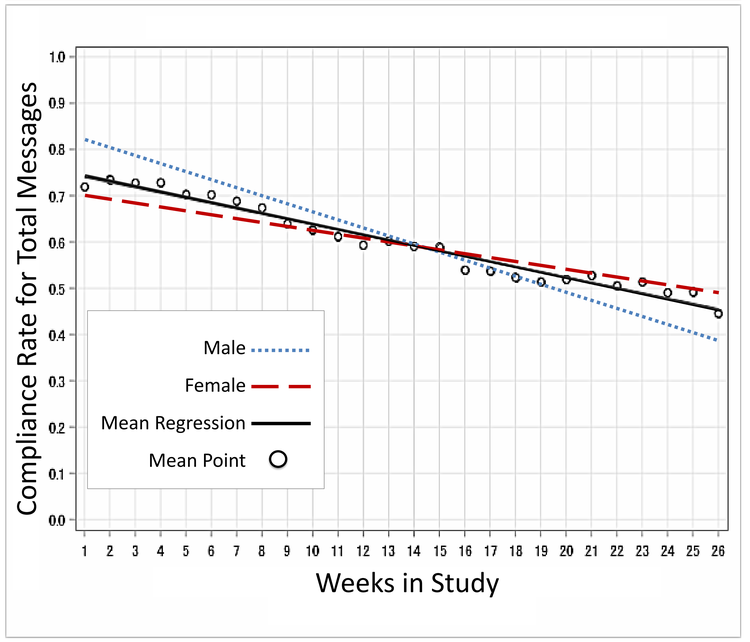


Figure 2. Linear Mixed Effects Model: Decline of Total Messages Viewed by Sex.

Conclusions

- Age was the sole predictor inversely related to aggregated intervention messages viewed across 182 days.
- Number of messages viewed progressively decreased across the 6-month study period, but the rate of decline varied across predictor variables.

Discussion

- Research that examines compliance across predictors may benefit from larger and more representative samples.
- Future smartphone-based interventions should use mixed methods approaches to identify barriers to viewing prompted messages.
- Strategies that mitigate message fatigue (e.g., tailored visual aids) may increase engagement and efficacy.

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